

Prostate cancer

Managed on two axes — **extent** (M0 vs M1) crossed with **castration status** (sensitive vs resistant). Castration-resistant = progression despite testosterone <50 ng/dL — **always confirm a castrate level before applying the label**

Stanford Oncology Handbook 2025, Prostate Cancer (Chien J, Parikh D, Srinivas S) · NCCN · items marked † are outside the handbook chapter

WORKUP

- Risk inputs: **cT stage, Grade Group, PSA**, number/% positive cores, PSA density
- Imaging by risk: **none** for very low / low / favorable intermediate · soft-tissue ± bone for unfavorable intermediate · **bone AND soft-tissue for high / very high**
- **Germline** (ATM, BRCA1/2, CHEK2, HOXB13, TP53): metastatic or node+, high/very-high risk, Ashkenazi ancestry, family history; consider if age ≤55
- **Somatic**: HRR in metastatic disease; MSI-H/dMMR and TMB in mCRPC

GRADE GROUP & RISK

Grade Group	GG1 ≤6 · GG2 3+4 · GG3 4+3 · GG4 = 8 · GG5 = 9–10
Very low / low	T1c + GG1 + PSA<10 + <3 cores ≤50% + PSAd <0.15 / cT1–T2a + GG1 + PSA <10
Favorable interm.	1 IRF + GG1–2 + <50% cores positive
Unfavorable interm.	2–3 IRFs, or GG3, or ≥50% cores positive
High	cT3–T4, or GG4–5, or PSA >20
Very high	≥2 of: cT3–T4, GG4–5, PSA >40

IRF (intermediate risk factor) = cT2b–cT2c, GG2 or 3, PSA 10–20.

ADT DURATION WITH RT — THE EVIDENCE

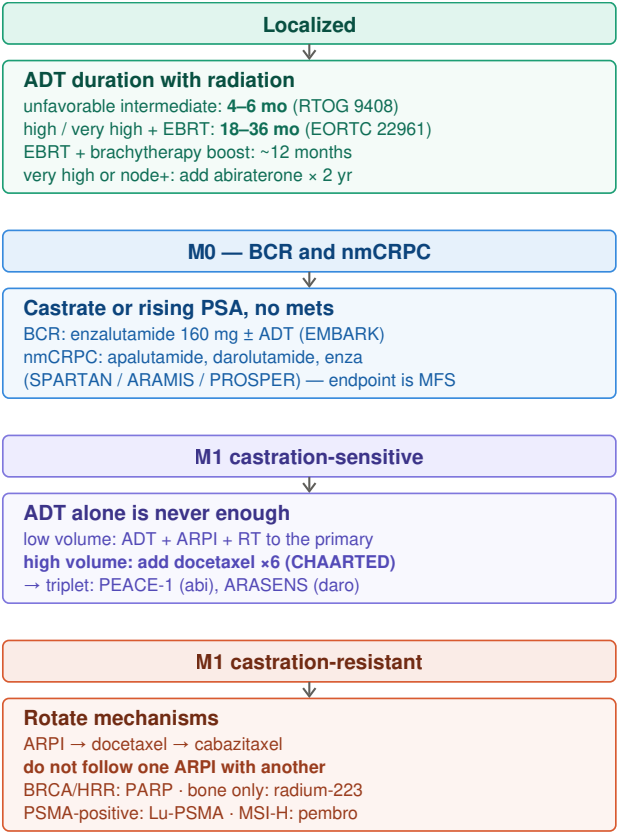
RTOG 9408	RT ± 4 mo ADT, mostly intermediate risk — OS + cause-specific benefit; established short course
EORTC 22863	RT ± 3 yr ADT, locally advanced — 10-yr OS 58% vs 40%
EORTC 22961 · RTOG 9202 · DART	6 mo vs 3 yr — 6 mo not non-inferior (5-yr OS 81% vs 85%); RTOG 9202 and DART likewise favor the long course, especially Gleason 8–10
PCS IV (Nabid)	18 vs 36 mo in high risk — 18 mo non-inferior
† JAMA Oncol 2026	IPD meta-analysis, 13 trials, n=10,266 — benefit is non-linear and plateaus after ~12 mo : intermediate <6 mo, high ~12 mo for most

PCS IV and the 2026 meta-analysis both argue the shorter end of 18–36 months is usually enough.

PEARLS

- **ADT continues indefinitely** — castration resistance means the AR pathway is still active, not abandoned
- † Rapid progression with a **disproportionately low PSA** → suspect neuroendocrine transformation; biopsy and consider platinum

TREATMENT PATHWAY



ADT AND HORMONAL AGENTS

Agent	Dosing	Schedule
Leuprolide	7.5 / 22.5 / 30 / 45 mg IM or SQ	q4 / 12 / 16 / 24 wk
Goserelin · triptorelin	3.8 or 10.6 mg IM · 3.75 or 11.25 mg IM	q4 / q12 wk
Degarelix (antagonist)	240 mg SQ load, then 80 mg SQ	q4 wk
Relugolix (oral antagonist)	360 mg PO load, then 120 mg PO (HERO)	daily
Bicalutamide	50 mg PO (150 mg high-dose)	daily
Abiraterone	1000 mg + prednisone 5 mg daily (M1 CSPC) or 5 mg BID (CRPC)	daily
Enzalutamide · apalutamide · darolutamide	160 mg daily · 240 mg daily · 600 mg BID	PO

LHRH agonists cause a **testosterone flare** — cover with bicalutamide started 7 days prior, or use an antagonist. Antagonists reach castrate levels in ~3 days. Low-cost option: abiraterone 250 mg with a low-fat breakfast.

CHEMOTHERAPY, RADIOPHARMACEUTICALS, IMMUNOTHERAPY

Agent	Dosing	Schedule
Docetaxel (M1 CSPC)	75 mg/m² IV — without prednisone, 6 doses (CHAARTED)	q3wk
Docetaxel (mCRPC)	75 mg/m² IV + prednisone 5 mg BID	q3wk
Cabazitaxel	20 or 25 mg/m² + prednisone (G-CSF if 25)	q3wk
Radium-223	50 kBq/kg IV × 6 — symptomatic bone mets, no visceral (ALSYMPCA)	q4wk
Lu-177-PSMA-617	7.4 GBq IV × 4–6 — PSMA-positive on PET (VISION)	q6wk
Sipuleucel-T	IV × 3 doses — minimally symptomatic mCRPC	q2wk

BIOMARKER → THERAPY

Alteration	Therapy
BRCA, 1L mCRPC	Niraparib 200 + abiraterone (MAGNITUDE) · olaparib 300 BID + abiraterone (PROpel)
HRR, 1L mCRPC	Talazoparib 0.5 mg + enzalutamide (TALAPRO-2)
HRR, post-ARPI	Olaparib 300 mg BID (PROfound) · rucaparib 600 mg BID (TRITON2)
MSI-H / dMMR / TMB >10	Pembrolizumab 200 mg q3wk

PARP **monotherapy** belongs in the post-ARPI setting; PARP + **ARPI combinations** are first-line.

† **Bone health**: baseline DEXA before starting long-term ADT; bone-protective agent for CRPC bone metastases. **Do not combine radium-223 with abiraterone** — ERA-223 showed more fractures and deaths.